

Laércio LIMA PILLA

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Researcher in High-Performance Computing

CURRICULUM VITAE: JUNE, 2025

Education

Univ. Grenoble Alpes & Federal University of Rio Grande do Sul <i>PhD in Computer Science</i> Thesis Advisers: Prof. Jean-François Méhaut & Prof. Philippe O. A. Navaux	FR & BR 2010–2014
Federal University of Rio Grande do Sul <i>Bachelor in Computer Science</i>	BR 2005–2009

Work Experience

CNRS <i>Research Scientist, UMR5800 - LaBRI, Topal team (prev. Storm team)</i>	Talence, FR 2021–
<i>Research Scientist, UMR8623 - Laboratoire de Recherche en Informatique, ParSys team</i>	Orsay, FR 2018–2020
Inria Grenoble - Rhône-Alpes <i>Postdoctoral Researcher</i>	Grenoble, FR 2018–2018
Federal University of Santa Catarina <i>Associate Professor</i>	Florianópolis, BR 2014–2017
LIG & Parallel and Distributed Processing Group <i>Assistant Researcher</i>	Grenoble, FR & Porto Alegre, BR 2010–2014
Parallel and Distributed Processing Group <i>Junior Researcher, Microsoft Interop-Lab UFRGS</i>	Porto Alegre, BR 2008–2009
Parallel and Distributed Processing Group <i>Junior Researcher</i>	Porto Alegre, BR 2007–2008

Honors & Awards

2025: Best Paper Award for “*Scheduling Strategies for Partially-Replicable Task Chains on Two Types of Resources*”, Heterogeneity in Computing Workshop (HCW)

2019: Best Paper Award for “*On server-side file access pattern matching*”, International Conference on High Performance Computing & Simulation (HPCS)

2017: 3rd Place Award on Multi-Deck Stand. Cell Legalization, ICCAD CAD Contest

2015: Best Paper Award for “*An Efficient Algorithm for Communication-Based Task Mapping*”, 23rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP)

2014: Best PhD Thesis in the Computer Architecture and High Performance Computing Contest (WSCAD-CTD)

2010: 2nd place, 5th International Marathon of Parallel Programming at SBAC-PAD

2009: 1st place, 4th International Marathon of Parallel Programming at SBAC-PAD

Diversity, Equity, and Inclusion

2023–: Member of the National Gender Equality and Equal Opportunities Committee at Inria

2022–: External member of the Equality, Diversity & Inclusion committee from the Computer Science Department of the University of Sheffield in the UK

2022–: Responsibility for the Gender Equality and Equal Opportunities workgroup in the Inria Center at the University of Bordeaux

2021–: Member of the Societal Challenges commission in LaBRI

Research Projects

2025–2028: Collaborative Unified Platform for a Scalable and Efficient Learning Infrastructure (CUPSELI), *Défi Commun Inria - hive*. Member

2024–2026: Data movement, Energy CONsumption and performance in High-Performance Computing (DECoHPC), *Inria + LNCC, UFF, UFRGS, and CEFET-RJ joint team*. Member

2022–2026: PUShing Low-carbon Services towards the Edge (PULSE), *Défi Commun Inria – Qarnot computing with the support of Ademe*. Member

2022–2024: Study and research into telecommunication chain optimization using AFF3CT software, *DGA/AID funding*. PI

2022–2024: Maelstrom, *Inria + Simula joint team*. Member

2021–2022: 3BEARS: Broad Bundle of BEnchmARks for Scheduling in HPC, Big Data, and ML, *PHC Germaine de Staël, (Project number 2021-13)*. PI

2019–2020: Exploring the Trade-offs of Scheduling in Two Contexts, *LRI Internal Funding*. PI

2019–2019: Mapping for Emerging Memory Models in HPC Systems, *CNRS INS2I PEPS*. PI

2017–2018: Adaptive Global Scheduling for Scientific Applications, *CNPq Universal (grant 401266/2016-8)*. PI

2016–2017: Energy-aware Scheduling and Fault Tolerance Techniques for the Exascale Era, *STIC-AmSud (grant 99999.007556/2015-02)*. Member

Supervisions

Post-doctoral Fellows:

2022–2024: Co-supervisor of Esragul Korkmaz, Inria

PhD Students (3-4y graduate program):

2025–: Co-advisor of Vanderlei Munhoz Pereira Filho, “*Scheduling of task-based parallel applications on heterogeneous Cloud computing environments*”, U. Bordeaux & UFSC with O. Aumage and M. Castro

2023–: Co-supervisor of Alan Lira Nunes, “*Scheduling algorithms for the optimization of distributed machine learning models on heterogeneous resources*”, U. Bordeaux & UFF with L. M. A. Drummond and C. Boeres

2022–: Co-advisor of Diane Orhan, “*Modeling and dynamic optimization of software radio chains on heterogeneous architectures*”, U. Bordeaux with C. Jégo and D. Barthou

Master's Students (2y graduate program):

2018–2020: Co-supervisor of Vinicius Marino Calvo Torres de Freitas, “*Diffusive load balancing for high performance computing applications: workload discretization and graph processing*”, UFSC

2017–2019: Co-supervisor of Sheiny Fabre Almeida, “*Accelerating the legalization of integrated circuits using partitioning with k-d trees*”, UFSC

2017–2019: Co-supervisor of Alexandre De Limas Santana, “*A design method for supporting the development and integration of ARTful global schedulers into multiple programming models*”, UFSC

2016–2018: Co-supervisor of Giovane de Oliveira Torres, “*An evaluation of memory controllers for non-volatile memories*”, UFPel

2015–2017: Co-supervisor of Vitor Alano de Ataídes, “*OpenStack Orchestra, a virtual machine scheduler and load balancer for OpenStack Clouds*”, UFPel

2014–2015: Co-supervisor of Rodrigo de Moura, “*Potential for trace reuse in ARM architectures*”, UFPel

Undergraduate Students and Interns:

2025–: Co-supervisor of Noureddine Tamssaout, M2 @ U. of Perpignan Via Domitia, Inria

2024–2025: Co-supervisor of Enrique Galvez, M2 @ ENS Lyon, Inria

2023–2023: Co-supervisor of Frédéric Becerril, L3 @ ENS Lyon, Inria

2023–2023: Co-supervisor of Angel Hippolyte, M1 @ U. Bordeaux, Inria

2022–2022: Co-supervisor of Thomas Morin, M1 @ U. Bordeaux, Inria

2022–2022: Co-supervisor of Pélagie Alves, M2 @ Bordeaux INP, Inria

2022–2022: Co-supervisor of Diane Orhan, M2 @ Bordeaux INP, Inria

2021–2021: Co-supervisor of Mustapha Regragui, M1 @ Bordeaux INP, Inria

2021–2021: Co-supervisor of Lana Scravaglieri, M1 @ ENSIIE, Inria

2017–2017: Co-supervisor of João Vítor Valente Tarouco de Oliveira, UFPel

2016–2016: Supervisor of Adrien Mouret, M1 @ Polytech Grenoble, UFSC

2015–2018: Supervisor of Vinicius Marino Calvo Torres de Freitas, UFSC

2015–2017: Co-supervisor of Sheiny Fabre Almeida, UFSC

2015–2015: Supervisor of Igor da Silva Solecki, UFSC

2015–2015: Co-supervisor of Bruno Ricardo Machado, UFSC

2015–2015: Supervisor of Youcef Hamdani, M1 @ Polytech Grenoble, UFSC

2013–2013: Co-supervisor of Tiago Covolan Bozzetti, UFRGS

2011–2011: Co-supervisor of Gennaro Severino Rodrigues, UFRGS

Publications and Scientific Content

Journals.....

- [J24] D. Orhan, L. L. Pilla, D. Barthou, A. Cassagne, O. Aumage, R. Tajan, C. Jégo, C. Leroux, “*Optimal scheduling algorithms for software-defined radio pipelined and replicated task chains on multicore architectures*”, Journal of Parallel and Distributed Computing, vol. 204, October 2025. <https://doi.org/10.1016/j.jpdc.2025.105106>
- [J23] L. Scravaglieri, L. L. Pilla, A. Guermouche, O. Aumage, E. Saillard, M. Popov, “*Optimizing Performance and Energy Across Problem Sizes Through a Search Space Exploration and Machine Learning*”, Journal of Parallel and Distributed Computing, vol. 180, October 2023. <https://doi.org/10.1016/j.jpdc.2023.104720>
- [J22] L. L. Pilla, “*Scheduling Algorithms for Federated Learning with Minimal Energy Consumption*”, in IEEE Transactions on Parallel and Distributed Systems, vol. 34, no. 4, April 2023. <https://doi.org/10.1109/TPDS.2023.3240833>

- [J21] M. Regragui, B. Coye, L. L. Pilla, R. Namyst, D. Barthou, "*SimSGamE : Scheduling simulator for modern game engines*", in Journal of Open Source Software, vol. 7, no. 76, pp. 4592, August 2022. <https://doi.org/10.21105/joss.04592>
- [J20] R. Netto, S. Fabre, T. A. Fontana, V. Livramento, L. L. Pilla, L. Behjat, J. L. Guntzel, "*Algorithm Selection Framework for Legalization Using Deep Convolutional Neural Networks and Transfer Learning*", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 41, no. 5, pp. 1481-1494, May 2022. <https://doi.org/10.1109/TCAD.2021.3079126>
- [J19] A. Santana, V. Freitas, M. Castro, L. L. Pilla and J.-F. Méhaut, "*ARTful: A specification for user-defined schedulers targeting multiple HPC runtime systems*", in Software: Practice and Experience, vol. 51, pp. 1622-1638. July 2021. <https://dx.doi.org/10.1002/spe.2977>
- [J18] V. Freitas, L. L. Pilla, A. Santana, M. Castro and J. Cohen, "*PackStealLB: A Scalable Distributed Load Balancer based on Work Stealing and Workload Discretization*", in Journal of Parallel and Distributed Computing, vol. 150, pp. 35-45, April 2021. <https://doi.org/10.1016/j.jpdc.2020.12.005>
- [J17] E. H. M. Cruz, M. Diener, L. L. Pilla, and P. O. A. Navaux, "*Online Thread and Data Mapping Using a Sharing-Aware Memory Management Unit*", in ACM Transactions on Modeling and Performance Evaluation of Computing Systems, vol. 5, no. 4, article 16, 28 pages, January 2021. <https://doi.org/10.1145/3433687>
- [J16] E. H. M. Cruz, M. Diener, L. L. Pilla, and P. O. A. Navaux, "*EagerMap: A Task Mapping Algorithm to Improve Communication and Load Balancing in Clusters of Multicore Systems*", in ACM Transactions on Parallel Computing, vol. 5, no. 4, article 17, 24 pages, March 2019. <https://doi.org/10.1145/3309711>
- [J15] M. Hinz, G. P. Koslovski, C. C. Miers, L. L. Pilla and M. A. Pillon, "*A Cost Model for IAAS Clouds based on Virtual Machine Energy Consumption*", in Journal of Grid Computing, vol. 16, no. 3, pp. 493-512, September 2018. <https://doi.org/10.1007/s10723-018-9440-8>
- [J14] A. S. Furtado, L. L. Pilla and V. Bogorny, "*A Branch and Bound Strategy for Fast Trajectory Similarity Measuring*", in Data and Knowledge Engineering, vol. 115, pp. 16-31, May 2018. <https://doi.org/10.1016/j.datak.2018.01.003>
- [J13] R. da R. Righi, R. de Q. Gomes, V. F. Rodrigues, C. A. da Costa, A. M. Alberti, L. L. Pilla and P. O. A. Navaux, "*MigPF: Towards on self-organizing process rescheduling of Bulk-Synchronous Parallel applications*", in Future Generation Computer Systems, vol. 78, part 1, pp. 272-286, January 2018. <https://doi.org/10.1016/j.future.2016.05.004>
- [J12] E. H. M. Cruz, M. Diener, L. L. Pilla and P. O. A. Navaux, "*Hardware-Assisted Thread and Data Mapping in Hierarchical Multicore Architectures*", in ACM Transactions on Architecture and Code Optimization, vol. 13, no. 3, article 28, 28 pages, September 2016. <https://doi.org/10.1145/2975587>
- [J11] E. H. M. Cruz, M. Diener, M. A. Z. Alves, L. L. Pilla and P. O. A. Navaux, "*LAPT: A locality-aware page table for thread and data mapping*", in Parallel Computing, vol. 54, pp. 59-71, May 2016. <https://doi.org/10.1016/j.parco.2015.12.001>
- [J10] D. A. G. de Oliveira, L. L. Pilla, T. Santini and P. Rech, "*Evaluation and Mitigation of Radiation-Induced Soft Errors in Graphics Processing Units*", in IEEE Transactions on Computers, vol. 65, no. 3, pp. 791-804, March 2016. <https://doi.org/10.1109/TC.2015.2444855>
- [J9] L. L. Pilla, D. A. G. Oliveira, C. Lunardi, P. O. A. Navaux, L. Carro and P. Rech, "*Memory Access Time and Input Size Effects on Parallel Processors Reliability*", in IEEE Transactions on Nuclear Science, vol. 62, no. 6, pp. 2627-2634, December 2015. <https://doi.org/10.1109/TNS.2015.2496381>

- [J8] L. L. Pilla, T. C. Bozzetti, M. Castro, P. O. A. Navaux and J.-F. Méhaut, “*ComprehensiveBench: a Benchmark for the Extensive Evaluation of Global Scheduling Algorithms*”, in Journal of Physics: Conference Series, vol. 649, conf. 1, 12 pages, October 2015. <https://doi.org/10.1088/1742-6596/649/1/012007>
- [J7] M. Diener, E. H. M. Cruz, L. L. Pilla, F. Dupros and P. O. A. Navaux, “*Characterizing communication and page usage of parallel applications for thread and data mapping*”, in Performance Evaluation, vol. 88-89, pp. 18–36, June 2015. <https://doi.org/10.1016/j.peva.2015.03.001>
- [J6] E. L. Padoin, L. L. Pilla, M. Castro, F. Z. Boito, P. O. A. Navaux and J.-F. Méhaut, “*Performance/energy trade-off in scientific computing: the case of ARM big.LITTLE and Intel Sandy Bridge*”, in IET Computers & Digital Techniques, vol. 9, no. 1, pp. 27–35, January 2015. <https://doi.org/10.1049/iet-cdt.2014.0074>
- [J5] L. L. Pilla, P. Rech, F. Silvestri, C. Frost, P. O. A. Navaux, M. Sonza Reorda and L. Carro, “*Software-Based Hardening Strategies for Neutron Sensitive FFT Algorithms on GPUs*”, in IEEE Transactions on Nuclear Science, vol. 61, no. 4, pp. 1874–1880, August 2014. <https://doi.org/10.1109/TNS.2014.2301768>
- [J4] L. L. Pilla, C. P. Ribeiro, P. Coucheney, F. Broquedis, B. Gaujal, P. O. A. Navaux and J.-F. Méhaut, “*A topology-aware load balancing algorithm for clustered hierarchical multi-core machines*”, in Future Generation Computer Systems, vol. 30, pp. 191–201, January 2014. [urlhttps://doi.org/10.1016/j.future.2013.06.023](https://doi.org/10.1016/j.future.2013.06.023)
- [J3] E. L. Padoin, L. L. Pilla, F. Z. Boito, R. V. Kassick, P. Velho and P. O. A. Navaux, “*Evaluating application performance and energy consumption on hybrid CPU+GPU architecture*”, in Cluster Computing, vol. 16, no. 3, pp. 511–525, September 2013. <https://doi.org/10.1007/s10586-012-0219-6>
- [J2] C. Osthoff, F. Z. Boito, R. V. Kassick, L. L. Pilla, P. O. A. Navaux, C. Schepke, J. Panetta, P. J. Grunmann, N. Maillard, P. L. S. Dias and R. L. Walko, “*Atmospheric models hybrid OpenMP/MPI implementation multicore cluster evaluation*”, in International Journal of Information Technology, Communications and Convergence, vol. 2, no. 3, pp. 212–233, November 2012. <http://doi.org/10.1504/ijitcc.2012.050411>
- [J1] R. da R. Righi, L. L. Pilla, N. Maillard, A. Carissimi and P. O. A. Navaux, “*Observing the impact of multiple metrics and runtime adaptations on BSP process rescheduling*”, in Parallel Processing Letters, vol. 20, no. 2, pp. 123–144, June 2010. <https://doi.org/10.1142/S0129626410000107>

Conferences.....

- [C40] A. Torri, P. Dominikowski, B. Pointal, O. Kaya, L. L. Pilla, O. Coulaud, “*Near-Optimal Contraction Strategies for the Scalar Product in the Tensor-Train Format*”, to be published in European Conference on Parallel Processing (Euro-Par), August 2025.
- [C39] D. Orhan, Y. Idouar, L. L. Pilla, A. Cassagne, D. Barthou, C. Jégo, “*Scheduling Strategies for Partially-Replicable Task Chains on Two Types of Resources*”, to be published in 2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW) – Heterogeneity in Computing Workshop (HCW), June 2025. <https://hal.science/hal-04941123>
- [C38] A. L. Nunes, C. Boeres, L. M. A. Drummond, L. L. Pilla, “*Optimal Time and Energy-Aware Client Selection Algorithms for Federated Learning on Heterogeneous Resources*”, in 2024 IEEE 36th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), November 2024. <https://doi.org/10.1109/SBAC-PAD63648.2024.00021>
- [C37] O. Beaumont, R. Bouzel, L. Eyraud-Dubois, E. Korkmaz, L. L. Pilla, A. van Kempen, “*A $1.25(1 + \epsilon)$ -Approximation Algorithm for Scheduling with Rejection Costs Proportional to Processing Times*”,

- in European Conference on Parallel Processing (Euro-Par), pp. 225–238, August 2024. https://doi.org/10.1007/978-3-031-69577-3_16
- [C36] M. Regragui, B. Coye, L. L. Pilla, R. Namyst, D. Barthou, “*Exploring scheduling algorithms for parallel task graphs: a modern game engine case study*”, in European Conference on Parallel Processing (Euro-Par), pp. 103–118, August 2022. https://doi.org/10.1007/978-3-031-12597-3_7
- [C35] L. L. Pilla, “*Optimal Task Assignment for Heterogeneous Federated Learning Devices*”, in 2021 IEEE International Parallel and Distributed Processing Symposium (IPDPS), pp. 661–670, May 2021. <https://doi.org/10.1109/IPDPS49936.2021.00074>
- [C34] J. H. M. Korndörfer, M. Bielert, L. L. Pilla and F. M. Ciorba, “*Mapping Matters: Application Process Mapping on 3-D Processor Topologies*”, to be published in the 2020 International Conference on High Performance Computing & Simulation (HPCS) 2021. <https://hal.archives-ouvertes.fr/hal-02617127v1>
- [C33] A. V. C. R. Oikawa, V. Freitas, M. Castro and L. L. Pilla, “*Adaptive Load Balancing based on Machine Learning for Iterative Parallel Applications*”, in 28th Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP), pp. 94–101, March 2020. <https://doi.org/10.1109/PDP50117.2020.00021>
- [C32] V. Freitas, A. Santana, M. Castro and L. L. Pilla, “*Distributed Memory Graph Representation for Load Balancing Data: Accelerating Data Structure Generation for Decentralized Scheduling*”, in International Conference on High Performance Computing & Simulation (HPCS), pp. 787–794, July 2019. <https://doi.org/10.1109/HPCS48598.2019.9188134>
- [C31] F. Z. Boito, R. Nou, L. L. Pilla, J. L. Bez, J.-F. Méhaut, T. Cortes and P. O. A. Navaux, “*On server-side file access pattern matching*”, in International Conference on High Performance Computing & Simulation (HPCS), pp. 217–224, July 2019. <https://doi.org/10.1109/HPCS48598.2019.9188092>
- [C30] R. Netto, S. Fabre, T. Fontana, V. Livramento, L. L. Pilla and J. L. Güntzel, “*How Deep Learning Can Drive Physical Synthesis Towards More Predictable Legalization*”, in Proceedings of the 2019 International Symposium on Physical Design (ISPD), pp 3–10, April 2019. <https://doi.org/10.1145/3299902.3309754>
- [C29] V. Freitas, A. L. Santana, M. Castro and L. L. Pilla, “*A Batch Task Migration Approach for Decentralized Global Rescheduling*”, in Proceedings of the 30th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), pp. 49–56, September 2018. <https://doi.org/10.1109/CAHPC.2018.8645953>
- [C28] A. Santana, V. Freitas, L. L. Pilla, M. Castro and J.-F. Méhaut, “*Reducing Global Schedulers’ Complexity Through Runtime System Decoupling*”, in Proceedings of the 2018 Symposium on High Performance Computing Systems (WSCAD), pp.38–44, September 2018. <https://doi.org/10.1109/WSCAD.2018.00016>
- [C27] S. Fabre, J. L. Güntzel, L. L. Pilla, R. Netto, T. Fontana and V. Livramento, “*Enhancing Multi-Threaded Legalization Through k-d Tree Circuit Partitioning*”, in Proceedings of the 31st Symposium on Integrated Circuits and Systems Design (SBCCI), pp. 1–6, August 2018. <https://doi.org/10.1109/SBCCI.2018.8533264>
- [C26] E. Cruz, M. Diener, M. Serpa, P. Navaux, L. L. Pilla and I. Koren, “*Improving Communication and Load Balancing with Thread Mapping in Manycore Systems*”, in 26th Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP), pp. 93–100, March 2018. <https://doi.org/10.1109/PDP2018.2018.00021>

- [C25] D. Oliveira, L. Pilla, N. DeBardeleben, S. Blanchard, H. Quinn, I. Koren, P. Navaux and Paolo Rech, "*Experimental and analytical study of Xeon Phi reliability*", in Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC), article 28, 12 pages, November 2017. <https://doi.org/10.1145/3126908.3126960>
- [C24] T. A. Fontana, S. Almeida, R. Netto, V. Livramento, C. Guth, L. Pilla and J. L. Güntzel, "*Exploiting cache locality to speedup register clustering*", in Proceedings of the 30th Symposium on Integrated Circuits and Systems Design: Chip on the Sands (SBCCI), pp. 191–197, September 2017. <https://doi.org/10.1145/3109984.3110005>
- [C23] T. Fontana, R. Netto, V. Livramento, C. Guth, S. Almeida, L. Pilla and J. L. Güntzel, "*How Game Engines Can Inspire EDA Tools Development: A use case for an open-source physical design library*", in Proceedings of the 2017 ACM on International Symposium on Physical Design (ISPD), pp. 25–31, March 2017. <https://doi.org/10.1145/3036669.3038248>
- [C22] D. A. G. de Oliveira, L. L. Pilla, M. Hanzich, V. Fratin, F. Fernandes, C. Lunardi, J. M. Cela, P. O. A. Navaux, L. Carro and P. Rech, "*Radiation-Induced Error Criticality in Modern HPC Parallel Accelerators*", in IEEE International Symposium on High Performance Computer Architecture (HPCA), pp. 577–588, February 2017. <https://doi.org/10.1109/HPCA.2017.41>
- [C21] R. Netto, C. Guth, V. Livramento, M. Castro, L. L. Pilla and J. L. Güntzel, "*Exploiting parallelism to speed up circuit legalization*", in IEEE International Conference on Electronics, Circuits and Systems (ICECS), pp. 624–627, December 2016. <https://doi.org/10.1109/ICECS.2016.7841279>
- [C20] R. C. De Moura, G. O. Torres, M. L. Pilla, L. L. Pilla, A. T. Da Costa and F. M. G. França, "*Value Reuse Potential in ARM Architectures*", in 28th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), pp. 174–181, October 2016. <https://doi.org/10.1109/SBAC-PAD.2016.30>
- [C19] E. L. Padoin, L. L. Pilla, M. Castro, P. O. A. Navaux and J.-F. Méhaut, "*Exploration of Load Balancing Thresholds to Save Energy on Iterative Applications*", in Third Latin American Conference in High Performance Computing (CARLA), pp. 76–88, September 2016. https://doi.org/10.1007/978-3-319-57972-6_6
- [C18] E. H. M. Cruz, M. Diener, L. L. Pilla and P. O. A. Navaux, "*A Sharing-Aware Memory Management Unit for Online Mapping in Multi-core Architectures*", in 22nd International Conference on Parallel and Distributed Computing (Euro-Par), pp. 490–501, August 2016. https://doi.org/10.1007/978-3-319-43659-3_36
- [C17] D. A. G. de Oliveira, L. Pilla, C. Lunardi, L. Carro, P. O. A. Navaux and Paolo Rech, "*The Path to Exascale: Code Optimizations and Hardening Solutions Reliability*", in Proceedings of the 5th Workshop on Fault Tolerance for HPC at eXtreme Scale (FTXS), pp. 55–62, June 2015. <https://doi.org/10.1145/2751504.2751510>
- [C16] E. H. M. Cruz, M. Diener, L. L. Pilla and P. O. A. Navaux, "*An Efficient Algorithm for Communication-Based Task Mapping*", in 23rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP), pp. 207–214, March 2015. <https://doi.org/10.1109/PDP.2015.25>
- [C15] E. L. Padoin, M. Castro, L. L. Pilla, P. O. A. Navaux and J. F. Méhaut, "*Saving energy by exploiting residual imbalances on iterative applications*", in 21st International Conference on High Performance Computing (HiPC), 10 pages, December 2014. <https://doi.org/10.1109/HiPC.2014.7116895>

- [C14] D. A. G. Oliveira, P. Rech, L. L. Pilla, P. O. A. Navaux and L. Carro, “*GPGPUs ECC efficiency and efficacy*”, in IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT), pp. 209–215, October 2014. <https://doi.org/10.1109/DFT.2014.6962085>
- [C13] E. H. M. Cruz, M. Diener, M. A. Z. Alves, L. L. Pilla and P. O. A. Navaux, “*Optimizing Memory Locality Using a Locality-Aware Page Table*”, in IEEE 26th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), pp. 198–205, October 2014. <https://doi.org/10.1109/SBAC-PAD.2014.22>
- [C12] D. A. G. Oliveira, C. B. Lunardi, L. L. Pilla, P. Rech, P. O. A. Navaux and L. Carro, “*Radiation Sensitivity of High Performance Computing Applications on Kepler-Based GPGPUs*”, in 44th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), pp. 732–737, June 2014. <https://doi.org/10.1109/DSN.2014.75>
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- [C10] R. K. Tesser, L. L. Pilla, F. Dupros, P. O. A. Navaux, J. F. Méhaut and C. Mendes, “*Improving the Performance of Seismic Wave Simulations with Dynamic Load Balancing*”, 22nd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP), pp. 196–203, February 2014. <https://doi.org/10.1109/PDP.2014.37>
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- [T1] L. L. Pilla, “*Basics of Vectorization for Fortran Applications*”, January 2018. <https://inria.hal.science/hal-01688488>

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- [B5] A. V. Schmidt, F. Z. Boito and L. L. Pilla, “*Fundamentos de Estatística para Análise de Desempenho*”, in Anais da XVII Escola Regional de Alto Desempenho (ERAD/RS), editors E. L. Padoin, M. C. Cera and A. S. Charao, vol. 17, chapter 1, pp. 3-24, Sociedade Brasileira de Computação, 2017.
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- [D4] O. Beaumont, L. Eyraud-Dubois, E. Korkmaz, L. L. Pilla, “*Experimental codes and results for the paper “A $5/4(1+\epsilon)$ -Approximation Algorithm for Scheduling with Rejection Costs Proportional to Processing Times”*”, March 2024. <https://inria.hal.science/hal-04517532/>

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- [D1] L. L. Pilla and V. Freitas, "*Performance results of LeanMD on the Joliot-Curie supercomputer using different load balancing algorithms*", June 2020. <https://doi.org/10.5281/zenodo.3878953>

Professional Service

PC Member: CARLA (2015–2019), COMPAS (2021–2023), DRMEC (2019), ESSA (2025), IC2E (2025), ICCS (2019–2022), HCW (2020–2023, 2025), HPCEuropeLatAm (2022), ISC PhD Forum (2019, 2021), PAW-ATM (2022), SBAC-PAD (2014, 2015, 2019, 2023), SBESC (2017), SC (2020), SC Tutorials (2020), WCC (2023)

Journal Reviewer: 4OR, ACM Surveys, ACM TACO, ACM TCPS, CCPE, FGCS, IEEE TC, IEEE TNS, IEEE TPDS Reproducibility, IEEE TVLSI, JPDC, JSS, PARCO, SPE

Juries

2024: Young researchers (CRCN & ISFP) recruitment campaign, Inria Centre at the University of Rennes

2023: Zhiyuan Yao's PhD defense, "*Intelligent Orchestration of High Performance, Scalable, and Correct Data Centres As a Network Service*", Institut Polytechnique de Paris

2020: Luís Felipe Garlet Milani's PhD defense, "*Autotuning with machine learning of OpenMP task applications*", Univ. Grenoble Alpes

2018: Arthur Francisco Lorenzon's PhD defense, "*Aurora: Seamless Optimization of OpenMP Applications*", UFRGS

2015: Emílio Wuerges's PhD defense, "*WCET-Aware Prefetching of Unlocked Instruction Caches: a Technique for Reconciling Real-Time Constraint and Energy Efficiency*", UFSC

2015: Aldelir Fernando Luiz's PhD defense, "*Protocolos tolerantes a faltas bizantinas para transações em bancos de dados*", UFSC

Teaching: Long Courses

Federal University of Santa Catarina, BR: Advanced Topics in Computer Architecture, Compilers Construction, Computer Architecture and Organization, Computer Organization Laboratory, Introduction to Informatics, Parallel Computing (graduate course).

INP Bordeaux, FR: Algorithms for High-Performance Computing Platforms (*currently teaching*).

Polytech Paris-Saclay, FR: GPU Programming.

Université Paris-Saclay, FR: Scheduling and Runtime Systems.

Teaching: Short Courses

2017: Adaptive MPI, Massively Parallel Programming Week, National Laboratory for Scientific Computing (LNCC), Brazil

2016: Adaptive MPI, Massively Parallel Programming Week, National Laboratory for Scientific Computing (LNCC), Brazil

2015: Programming with Parallel Objects: from MPI to Charm++, 27th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)

2015: Parallel Programming with Charm++, 16th High Performance Computing Regional School

Event Organization

2022: IEEE 34th International Symposium on Computer Architecture and High Performance Computing, Organizing Committee & EDI Chair

2015: IEEE 27th International Symposium on Computer Architecture and High Performance Computing, Organizing Committee